Nivel 1

Ejercicio 1

Su tarea es diseñar y crear una tabla llamada “credit_card” que almacene detalles cruciales sobre las tarjetas de crédito. La nueva tabla debe poder identificar de forma única cada tarjeta y establecer una relación adecuada con las otras dos tablas ("transacción" y "empresa"). Después de crear la tabla será necesario que ingrese la información en el documento llamado "data_introduct_credit". Recuerda mostrar el diagrama y hacer una breve descripción del mismo.

Se ha creado la tabla credit_card con las columnas id, iban, pan, pin, cvv y expiring_date. La columna id está establecida como llave primaria.

```sql
USE transactions;
CREATE TABLE IF NOT EXISTS credit_card (
    id VARCHAR(15) PRIMARY KEY,
    iban VARCHAR(50),
    pan VARCHAR(20),
    pin VARCHAR(4),
    cvv INT,
    expiring_date VARCHAR(10)
    );
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	21:39:02	USE transactions	0 row(s) affected	0.000 sec
2	21:39:12	CREATE TABLE IF NOT EXISTS credit_card (id VARCHAR(15) PRIMARY KEY, iban VARCHAR(50), pan VARC...	0 row(s) affected	0.031 sec

```sql
-- Insertamos datos de credit_card
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-2938', 'TR301950312213576817638661', '5424465566813633', '3257', '984', '10/30/22');
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-2945', 'DO26854763748537475216568689', '5142423821948828', '9080', '887', '08/24/23');
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-2952', 'BG45IVQL52710525608255', '4556 453 55 5287', '4598', '438', '06/29/21');
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-2959', 'CR7242477244335841535', '372461377349375', '3583', '667', '02/24/23');
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-2966', 'BG72LKTQ15627628377363', '448566 886747 7265', '4900', '130', '10/29/24');
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-2973', 'PT8780622813509242945634 6', '544 58654 54343 384', '8760', '887', '01/30/25');
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-2980', 'DE39241881883086277136', '402400 7145845969', '5075', '596', '07/24/22');
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-2987', 'GE896814348377487818 13', '3763 747687 76666', '2298', '797', '10/31/23');
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-2994', 'BH627144283680667652 94', '344283273252593', '7545', '595', '02/28/22');
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-3001', 'CY490874266547745812668321 10', '511722 924833 2244', '9562', '867', '09/16/22');
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-3008', 'LU507216693616119230', '4485744464433884', '1856', '740', '04/05/25');
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-3015', 'PS1193982162957159683424568 21', '3784 662233 17389', '3246', '822', '01/31/22');
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-3022', 'GT91695162850556977423121857', '5164 1379 4842 3951', '5610', '342', '04/25/25');
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-3029', 'AZ62317413982441418123739746', '3429 279566 77631', '9708', '505', '09/02/23');
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-3036', 'AZ39336002925842865843941994', '3768 451556 48766', '2232', '565', '10/27/25');
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-3043', 'TN648814331051485217953 5', '455676 6437463635', '5969', '196', '06/07/25');
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-3050', 'FR5167744369175836831854477', '4024007123722', '4834', '126', '10/09/23');
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-3057', 'LU931822574697545215', '3484 621767 21237', '6805', '848', '09/14/25');
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-3064', 'PS1469655454492533776272731 33', '3467 732741 26810', '3865', '498', '06/03/25');
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-3071', 'NO8923814763512', '3464 789562 23352', '6625', '661', '12/20/23');
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-3078', 'IS0251271458846232795487 33', '4539 322 74 2377', '9405', '720', '03/08/23');
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-3085', 'BE63114723972437', '5266 3346 1135 1687', '7241', '413', '05/10/23');
INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-3092', 'RO65LSOD1166122125447487', '3488 754223 46253', '9417', '594', '12/19/22');
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
5004	21:48:40	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9580', 'XX78125888985195080...	1 row(s) affected	0.000 sec
5005	21:48:40	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9581', 'XX91567051640538812...	1 row(s) affected	0.000 sec
5006	21:48:46	SELECT * FROM transactions.credit_card LIMIT 0, 2000	2000 row(s) returned	0.000 sec / 0.000 sec

Se ha creado la foreign key fk_transaction_credit_card en la tabla transaction para establecer una relación de n a 1, esto se hizo deshabilitando los foreign key checks para evitar un error 1452 y porque estamos en un entorno controlado y corriendo en local, si fueran otras circunstancias, alteraríamos la columna credit_card_id de la tabla transaction para eliminar la reference a credit_card_id (la referencia estaba en la estructura de datos original antes de que credit_card se crease).

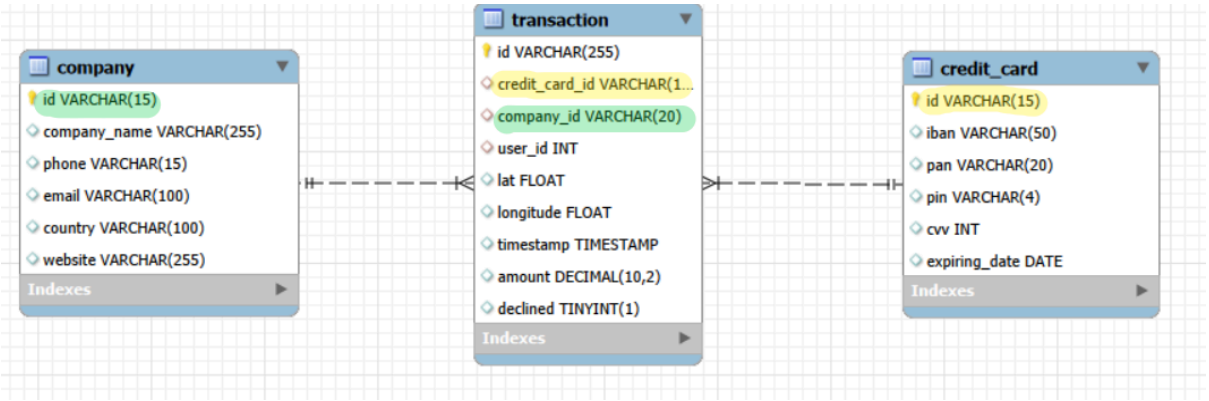
```
-- Desabilito los foreign key checks porque sino me da error (por el reference que tiene transaction.credit_card_id en su código de creación (Error Code 1452)
SET FOREIGN_KEY_CHECKS = 0;

-- Agrego el constraint que establece la foreign key de transaction a credit_card (relación n to 1)
ALTER TABLE transaction
ADD CONSTRAINT fk_transaction_credit_card
FOREIGN KEY (credit_card_id)
REFERENCES credit_card(id);

-- Vuelvo a activar los foreign key checks
SET FOREIGN_KEY_CHECKS = 1;
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
3	21:40:45	SET FOREIGN_KEY_CHECKS = 0	0 row(s) affected	0.000 sec
4	21:40:49	ALTER TABLE transaction ADD CONSTRAINT fk_transaction_credit_card FOREIGN KEY (credit_card_id) REFE...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.922 sec
5	21:41:14	SET FOREIGN_KEY_CHECKS = 1	0 row(s) affected	0.000 sec

El diagrama resultante:



También se hicieron modificaciones en la columna expiring_date (originalmente un VARCHAR) después de cargar los datos para que estos tuvieran el formato DATE correcto.

```
ALTER TABLE credit_card
ADD COLUMN expiring_date_2 DATE;

UPDATE credit_card
SET expiring_date_2 = str_to_date(expiring_date,'%m/%d/%y');

ALTER TABLE credit_card
DROP COLUMN expiring_date,
CHANGE COLUMN expiring_date_2 expiring_date DATE;
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
5007	21:49:51	ALTER TABLE credit_card ADD COLUMN expiring_date_2 DATE	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.032 sec
5008	21:49:58	UPDATE credit_card SET expiring_date_2 = str_to_date(expiring_date,'%m/%d/%y')	5000 row(s) affected Rows matched: 5000 Changed: 5000 Warnings: 0	0.204 sec
5009	21:50:03	ALTER TABLE credit_card DROP COLUMN expiring_date, CHANGE COLUMN expiring_date_2 expiring_date D...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec

Ejercicio 2

El Departamento de Recursos Humanos ha identificado un error en el número de cuenta asociado a la tarjeta de crédito con ID CcU-2938. La información que se mostrará para este registro es: TR323456312213576817699999. Recuerde demostrar que el cambio se realizó.

Registro antes de la transformación:

```
SELECT * FROM transactions.credit_card
WHERE id = 'CcU-2938';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

id	iban	pan	pin	cvv	expiring_date
CcU-2938	TR301950312213576817638661	5424465566813633	3257	984	2022-10-30
NULL	NULL	NULL	NULL	NULL	NULL

Código de la transformación:

```
UPDATE credit_card
SET iban = 'TR3234563122135768176999999'
WHERE id = 'CcU-2938';
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
5012	21:58:38	UPDATE credit_card SET iban = 'TR3234563122135768176999999' WHERE id = 'CcU-2938'	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec

Registro después de la transformación:

```
SELECT * FROM transactions.credit_card
WHERE id = 'CcU-2938';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

id	iban	pan	pin	cvv	expiring_date
CcU-2938	TR3234563122135768176999999	5424465566813633	3257	984	2022-10-30
NULL	NULL	NULL	NULL	NULL	NULL

Ejercicio 3

En la tabla "transacción" usted realiza una nueva transacción con la siguiente información:

Id: 108B1D1D-5B23-A76C-55EF-C568E49A99DD
credit_card_id: CcU-9999
company_id: b-9999
user_id: 9999
lat: 829.999
longitude: -117.999
amount: 111.11
declined: 0

Antes de la inserción, se han creado dos triggers, en credit_card y en company para automatizar la inserción de sus valores en dichas tablas y así evitar un error.

```
-- Trigger para insertar company_id si no existe
DELIMITER //
CREATE TRIGGER tg_company_before_transaction
BEFORE INSERT ON transaction
FOR EACH ROW
BEGIN
    IF NOT EXISTS (SELECT 1 FROM company WHERE id = NEW.company_id) THEN
        INSERT INTO company (id) VALUES (NEW.company_id);
    END IF;
END;
//
DELIMITER ;

-- Trigger para insertar credit_card_id si no existe
DELIMITER //
CREATE TRIGGER tg_credit_card_before_transaction
BEFORE INSERT ON transaction
FOR EACH ROW
BEGIN
    IF NOT EXISTS (SELECT 1 FROM credit_card WHERE id = NEW.credit_card_id) THEN
        INSERT INTO credit_card (id) VALUES (NEW.credit_card_id);
    END IF;
END;
//
DELIMITER ;
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
8	01:01:20	CREATE TRIGGER tg_company_before_transaction BEFORE INSERT ON transaction FOR EACH ROW BEGI...	0 row(s) affected	0.016 sec
9	01:01:40	CREATE TRIGGER tg_credit_card_before_transaction BEFORE INSERT ON transaction FOR EACH ROW BEG...	0 row(s) affected	0.015 sec

A continuación ingresamos el registro en transaction:

```
INSERT INTO tracompanynsaction (id,credit_card_id,company_id,user_id,lat,longitude,amount,declined)
VALUES ('108B1D1D-5B23-A76C-55EF-C568E49A99DD','CcU-9999','b-9999','9999','829.999','-117.999','111.11','0');
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
12	01:06:58	INSERT INTO transaction (id,credit_card_id,company_id,user_id,lat,longitude,amount,declined) VALUES ('108B...	1 row(s) affected	0.016 sec

Ejercicio 4

Desde recursos humanos se le solicita que elimine la columna "pan" de la tabla credit_card. Recuerda mostrar el cambio realizado.

Tabla antes de la transformación:

```
SELECT * FROM transactions.credit_card;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

id	iban	pan	pin	cvv	expiring_date
CcS-4857	XX4857591835292505850771	2314242385113924	1819	467	2025-09-27
CcS-4858	XX8581768137002436094025	6582720299715533	3964	817	2028-12-28
CcS-4859	XX7826930491423553609370	8861684536289642	4983	277	2026-11-26
CcS-4860	XX5559590368835304645299	2481155515498459	6876	661	2027-07-27
CcS-4861	XX2035182877195191627307	1308930301149557	5710	398	2026-04-25
CcS-4862	XX4774721462463645409758	6715617009807829	4042	174	2026-11-27
CcS-4863	XX1476829664245046207111	3140879819451394	5969	449	2029-12-27
CcS-4864	XX8380298893385731196159	5793672133649114	8481	139	2026-02-28
CcS-4865	XX7085078596101025280599	5101552687251312	7847	903	2028-11-25

Código de la transformación:

```
ALTER TABLE credit_card
DROP COLUMN pan;
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
5020	22:14:15	ALTER TABLE credit_card DROP COLUMN pan	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec

Tabla después de la transformación:

```
SELECT * FROM transactions.credit_card;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

id	iban	pin	cvv	expiring_date
CcS-4857	XX4857591835292505850771	1819	467	2025-09-27
CcS-4858	XX8581768137002436094025	3964	817	2028-12-28
CcS-4859	XX7826930491423553609370	4983	277	2026-11-26
CcS-4860	XX5559590368835304645299	6876	661	2027-07-27
CcS-4861	XX2035182877195191627307	5710	398	2026-04-25
CcS-4862	XX4774721462463645409758	4042	174	2026-11-27
CcS-4863	XX1476829664245046207111	5969	449	2029-12-27
CcS-4864	XX8380298893385731196159	8481	139	2026-02-28
CcS-4865	XX7085078596101025280599	7847	903	2028-11-25
CcS-4866	XX4792859188206596406839	9271	961	2025-02-28

Nivel 2

Ejercicio 1

Elimine de la tabla de transacciones el registro con ID 000447FE-B650-4DCF-85DE-C7ED0EE1CAAD de la base de datos.

```
DELETE FROM transaction
WHERE id = '000447FE-B650-4DCF-85DE-C7ED0EE1CAAD';
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
5024	22:19:39	DELETE FROM transaction WHERE id = '000447FE-B650-4DCF-85DE-C7ED0EE1CAAD'	1 row(s) affected	0.015 sec

Ejercicio 2

La sección de marketing desea tener acceso a información específica para realizar análisis y estrategias efectivas. Se ha solicitado una opinión para proporcionar detalles clave sobre las empresas y sus transacciones. Necesitará crear una vista llamada VistaMarketing que contenga la siguiente información: Nombre de la empresa. Teléfono de contacto. País de residencia. Compra media realizada por cada empresa. Presenta la vista creada, ordenando datos de mayor a menor compra promedio.

```
CREATE VIEW VistaMarketing AS
SELECT t.company_id,
       c.company_name,
       c.phone, c.country,
       ROUND(AVG(t.amount),2) AS average_amount_company
  FROM company c
  JOIN transaction t
    ON c.id=t.company_id
 GROUP BY t.company_id
 ORDER BY average_amount_company DESC;
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
5026	22:23:17	CREATE VIEW VistaMarketing AS SELECT t.company_id, c.company_name, c.phone, c.country, ROUND(AVG(t.amount),2) AS average_amount_company FROM c...	0 row(s) affected	0.016 sec

Se comprueba que se muestran correctamente:

```
SELECT *
  FROM VistaMarketing;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

company_id	company_name	phone	country	average_amount_company
b-2222	Ac Fermentum Incorporated	06 85 56 52 33	Germany	284.87
b-2282	Pretium Neque Corp.	07 77 48 55 28	Australia	276.16
b-2422	Urna Convallis Associates	06 01 24 77 04	United States	274.24
b-2538	At Associates	09 56 61 10 65	New Zealand	272.21
b-2498	Metus Vitae Associates	08 25 44 40 66	Australia	270.08
b-2570	Aliquet Diam Limited	02 76 61 47 46	United States	269.60
b-2470	Nec Luctus LLC	02 14 71 75 73	Norway	268.60
b-2382	Neque Tellus Incorporated	04 43 18 34 19	Ireland	267.85
b-2394	Tortor Nunc Commodo Company	05 35 92 77 16	United States	267.84
b-2514	Cras Consulting	07 50 10 85 63	Belgium	267.44
b-2398	Eget Ipsum Ltd	03 67 44 56 72	United States	267.06
b-2274	Sed LLC	01 63 16 26 52	Belgium	266.07
b-2334	Amet Institute	06 33 40 21 33	Australia	265.54
b-2462	Pede Ultrices Ltd	04 52 40 26 17	Belgium	265.36
b-2230	Fusce Corp.	08 14 97 58 85	United States	265.23
b-2486	Tincidunt Orci Limited	01 78 18 81 44	Norway	265.10
b-2242	Donec Ltd	01 25 51 37 37	Norway	265.01
b-2258	Vestibulum Lorem PC	02 02 87 33 40	Belgium	264.96
b-2574	Orci Adipiscing Limited	03 18 00 77 81	United Kingdom	264.79
b-2346	Ut Semper Foundation	01 60 36 33 06	Sweden	264.71

staMarketing 1 | Read On

Output

Action Output

#	Time	Action	Message	Duration / Fetch
5029	22:26:35	SELECT * FROM VistaMarketing LIMIT 0, 2000	101 row(s) returned	0.359 sec / 0.000 sec

Ejercicio 3

Filtre la vista VistaMarketing para mostrar solo las empresas que tienen su país de residencia en "Alemania"

```
SELECT *
  FROM VistaMarketing
 WHERE country = "Germany";
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

company_id	company_name	phone	country	average_amount_company
b-2222	Ac Fermentum Incorporated	06 85 56 52 33	Germany	284.87
b-2302	Nunc Interdum Incorporated	05 18 15 48 13	Germany	259.32
b-2234	Convallis In Incorporated	06 66 57 29 50	Germany	257.75
b-2358	Ac Industries	09 34 65 40 60	Germany	255.15
b-2614	Rutrum Non Inc.	02 66 31 61 09	Germany	255.14
b-2550	Auctor Mauris Corp.	05 62 87 14 41	Germany	254.77
b-2306	Augue Foundation	06 88 43 15 63	Germany	253.51
b-2566	Aliquam PC	01 45 73 52 16	Germany	253.14

VistaMarketing 2 × Read On

Output

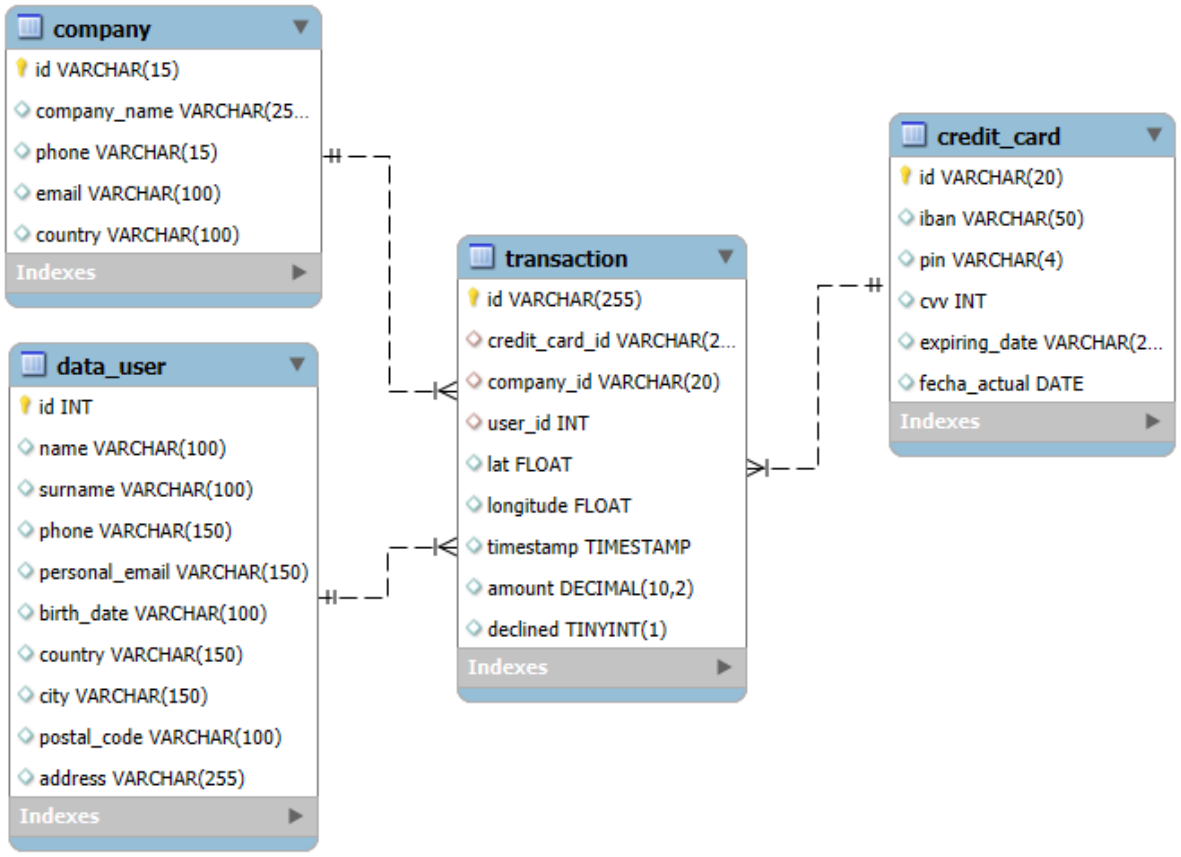
Action Output

#	Time	Action	Message	Duration / Fetch
5031	22:28:54	SELECT * FROM VistaMarketing WHERE country = "Germany" LIMIT 0, 2000	8 row(s) returned	0.343 sec / 0.000 sec

Nivel 3

Ejercicio 1

La próxima semana tendrás una nueva reunión con responsables de marketing. Un colega de su equipo hizo modificaciones en la base de datos, pero no recuerda cómo las hizo. Te pide que le ayudes a dejar los comandos ejecutados para obtener el siguiente diagrama:



Recordatorio

En esta actividad es necesario describir el “paso a paso” de las tareas realizadas. Es importante hacer descripciones sencillas, sencillas y fáciles de entender. Para realizar esta actividad tendrás que trabajar con los archivos denominados "structure_dades_user" y "data_introduct_user"

Recuerda seguir trabajando en el modelo y tablas con las que ya has trabajado hasta ahora.

Se crea la tabla user y se definen los atributos de sus campos según el archivo "estructura_dades_user"

```
CREATE TABLE IF NOT EXISTS user (
    id CHAR(10) PRIMARY KEY,
    name VARCHAR(100),
    surname VARCHAR(100),
    phone VARCHAR(150),
    email VARCHAR(150),
    birth_date VARCHAR(100),
    country VARCHAR(150),
    city VARCHAR(150),
    postal_code VARCHAR(100),
    address VARCHAR(255)
);
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
5033	22:41:27	CREATE TABLE IF NOT EXISTS user (id CHAR(10) PRIMARY KEY, name VARCHAR(100), surname VARCHAR(100), phone VARCHAR(150), email VARCHAR(150), birth_date V...	0 row(s) affected	0.032 sec

Se corrige el atributo de la columna id de la tabla user para que su tipo coincida con el de la tabla transaction (cambio de CHAR(10) a INT):

```
ALTER TABLE user
MODIFY COLUMN id INT;
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
5034	22:41:50	ALTER TABLE user MODIFY COLUMN id INT	0 row(s) affected Records: 0 Duplicates: 0 W...	0.047 sec

Se crea la foreign key fk_transaction_user:

```
SET FOREIGN_KEY_CHECKS = 0;
ALTER TABLE transaction
ADD CONSTRAINT fk_transaction_user
FOREIGN KEY (user_id)
REFERENCES user(id);
SET FOREIGN_KEY_CHECKS = 1;
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
5036	22:45:02	SET FOREIGN_KEY_CHECKS = 0	0 row(s) affected	0.000 sec
5037	22:45:07	ALTER TABLE transaction ADD CONSTRAINT fk_transaction_user FOREIGN KEY (user_id) REFERENCES user(id)	0 row(s) affected Records: 0 Duplicates: 0 W...	0.625 sec
5038	22:45:11	SET FOREIGN_KEY_CHECKS = 1	0 row(s) affected	0.000 sec

Se cargan los datos en la tabla user:

```
1  INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "151", "Meghan", "Hayden", "0800 746 6747", "arcu.vel@hotmail.c
2  INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "152", "Hakeem", "Alford", "(0111) 367 0184", "adipiscing.ligula
3  INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "153", "Keegan", "Pugh", "(016977) 3851", "sodales.nisi@aol.org"
4  INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "154", "Cooper", "Bullock", "(021) 2521 6627", "et@outlook.net",
5  INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "155", "Joshua", "Russell", "055 4409 5286", "justo.nec.ante@out
6  INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "156", "Remedios", "Case", "055 3114 1566", "mollis.phasellus.l
7  INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "157", "Philip", "Carey", "0800 640 6251", "phasellus@yahoo.net"
8  INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "158", "Fatima", "Dyer", "0800 1111", "adipiscing@google.org",
9  INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "159", "Kylynn", "Acevedo", "056 5727 9602", "dignissim.lacus.a
10 INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "160", "Lael", "Moody", "07123 850737", "nunc.sed@aol.org", "No
11 INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "161", "Nora", "Reeves", "(01692) 29410", "ac@aol.com", "Dec 24,
12 INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "162", "Francesca", "Sawyer", "(01632) 32279", "nunc@protonmail.
13 INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "163", "Denton", "Blackburn", "(015406) 66385", "tincidunt@outl
14 INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "164", "Preston", "Hood", "0845 46 47", "convallis.est.vitae@ya
15 INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "165", "Nora", "Cantrell", "0500 767716", "varius.et.euismod@pr
16 INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "166", "Matthew", "Woodard", "070 6461 3432", "urna@outlook.net"
17 INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "167", "Sheila", "Dickerson", "(01645) 68676", "et.lacinia@hotm
18 INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "168", "Donna", "Rivers", "07553 472381", "elit.erat@aol.org",
19 INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "169", "Linda", "Gaines", "(011370) 15148", "faucibus@outlook.e
20 INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "170", "William", "Benjamin", "0800 1111", "porttitor@icloud.ca"
21 INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "171", "Gary", "Robbins", "0500 856738", "dolor.dolor@google.ed
22 INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "172", "Yoko", "Calhoun", "055 0680 3951", "dui.nec@outlook.org"
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
10036	22:49:21	INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ("4980", "Asgaxi", "Btardzti", "+39-155-6819", "asgaxi.btardzti@...	1 row(s) affected	0.000 sec
10037	22:49:21	INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ("4983", "Eremdc", "Orekppbr", "+59-545-4710", "eremdc.orekp...	1 row(s) affected	0.000 sec
10038	22:49:21	INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ("4999", "Omjnoy", "Rmuqvxgw", "+32-354-7682", "omjnoy.rmuq...	1 row(s) affected	0.000 sec

Por consistencia, se crea el trigger para actualizar la tabla user cuando se actualice transaction:

```
188  DELIMITER //
189  CREATE TRIGGER tg_user_before_transaction
190  BEFORE INSERT ON transaction
191  FOR EACH ROW
192  BEGIN
193      IF NOT EXISTS (SELECT 1 FROM user WHERE id = NEW.user_id) THEN
194          INSERT INTO user (id) VALUES (NEW.user_id);
195      END IF;
196  END;
197  //
198  DELIMITER ;
```

Se comprueba que se ha realizado correctamente:

```
1  SELECT * FROM transactions.user;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

id	name	surname	phone	email	birth_date	country	city	postal_code	address
1	Zeus	Gamble	1-282-581-0551	interdum.enim@protonmail.edu	Nov 17, 1985	United States	New York	10001	348-7818 Sagittis St.
2	Garrett	Mcconnell	(718) 257-2412	integer.vitae.nibh@protonmail.org	Aug 23, 1992	United States	Philadelphia	19101	903 Sit Ave
3	Ciaran	Harrison	(522) 598-1365	interdum.feugiat@aol.org	Apr 29, 1998	United States	Houston	77001	736-2063 Tellus St.
4	Howard	Stafford	1-411-740-3269	ornare.egestas@icloud.edu	Feb 18, 1989	United States	Phoenix	85001	Ap #545-2244 Erat. Rd.
5	Hayfa	Pierce	1-554-541-2077	et.malesuada.fames@hotmail.org	Sep 26, 1998	United States	Philadelphia	19101	341-2821 Ultrices Av.
6	Joel	Tyson	(718) 288-8020	gravida.nunc.sed@yahoo.ca	Oct 15, 1989	United States	San Jose	95101	888-2799 Amet Street
7	Rafael	Jimenez	(817) 689-0478	eget@outlook.ca	Dec 4, 1981	United States	Chicago	60601	8627 Malesuada Rd.
8	Nissim	Franks	(692) 157-3469	egestas.aliquam.fringilla@google.ca	Aug 1, 1993	United States	New York	10001	Ap #251-7144 Integer St.
9	Mannix	Mcclain	(590) 883-2184	aliquam.nisl@outlook.com	Jan 24, 1987	United States	San Antonio	78201	647-3080 Lacus. St.
10	Robert	Mccarthy	(324) 746-6771	fermentum@protonmail.com	Apr 30, 1984	United States	San Jose	95101	P.O. Box 773
11	Joan	Baird	(981) 429-8106	et@outlook.net	Feb 25, 1990	United States	Los Angeles	90001	P.O. Box 687
12	Benedict	Wheeler	1-515-824-2855	tincidunt.donec.vitae@hotmail.couk	Aug 6, 1999	United States	Phoenix	85001	748-8694 Porttitor Avenue
13	Allegra	Stanton	1-927-753-6488	proin.eget@protonmail.ca	May 19, 1990	United States	New York	10001	4457 Ante. Av.
14	Sara	Flynn	1-311-646-9333	integer@outlook.net	Dec 27, 1988	United States	Los Angeles	90001	P.O. Box 865
15	Noelani	Patrick	1-723-488-5894	sem.magna@google.com	Sep 17, 1993	United States	Los Angeles	90001	596-5044 Sapien
16	Eric	Roth	1-218-549-8253	lorem.sit@yahoo.net	Sep 7, 1988	United States	San Diego	92101	P.O. Box 541
17	Bruce	Gill	(744) 732-8628	metus@aol.net	Mar 4, 1990	United States	Houston	77001	Ap #836-9508 Vitae

user 1 | Apply | Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
10037	22:49:21	INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ("4983", "Eremdc", "Orekppbr", "+59-545-4710", "eremdc.orek...	1 row(s) affected	0.000 sec
10038	22:49:21	INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ("4999", "Omjnoy", "Rmuqvxgw", "+32-354-7682", "omjnoy.rm...	1 row(s) affected	0.000 sec
10039	22:50:34	SELECT * FROM transactions.user LIMIT 0, 2000	2000 row(s) returned	0.000 sec / 0.000 sec

Ejercicio 2

La empresa también le pide que cree una vista llamada "ReportTecnico" que contenga la siguiente información:

ID de transacción
Nombre de usuario/aria
Apellido de usuario / aria
IBAN de la tarjeta de crédito utilizada.
Nombre de la empresa de la transacción realizada.
Asegúrese de incluir información relevante de las tablas que conocerá y utilice alias para cambiar el nombre de las columnas según sea necesario.

Muestra los resultados de la vista, ordene los resultados en orden descendente según la variable ID de transacción.

```
212     CREATE VIEW InformeTecnico AS
213     SELECT t.id AS id_transaction,
214            u.name AS user_name,
215            u.surname AS user_surname,
216            c_c.iban,
217            c.company_name
218       FROM transaction t
219       JOIN company c
220         ON t.company_id=c.id
221       JOIN credit_card c_c
222         ON t.credit_card_id=c_c.id
223       JOIN user u
224         ON t.user_id=u.id;
225
226     SELECT *
227       FROM InformeTecnico
228      ORDER BY id_transaction DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

id_transaction	user_name	user_surname	iban	company_name
FFFD31D6-9495-47CE-B54A-7DB8E1CC274B	Bmrgli	Tprvvmrc	XX7948144512112891824909 22	Turpis Company
FFFCF76D-ECF0-4985-A2D0-B2A7B75998FC	Dfrled	Vilqcjdl	XX6362517016478920366760 34	Amet Nulla Donec Corporation
FFFC9E8D-27C7-4ADE-98F2-7533EF4DF126	Securp	Faofvqfy	XX1626771433042236314375 67	Nunc Interdum Incorporated
FFFB270D-F53A-4D5D-9666-E5307C53CC84	Ggzjpa	Uirzjulh	XX3951142670820199525670 52	Viverra Donec Foundation
FFF9E3CE-234E-408C-A8EF-F9CAD577224A	Yshimq	Zpsjsleed	XX8845462156537570367941	Convallis In Incorporated
FFF9E178-6CD2-4DF9-99B0-49AE068809B1	Jevepx	Xwcwzwnm	XX3214055157116543847114 81	Mus Aenean Eget Foundation
FFF867C9-17B5-4B1F-AFD9-F8023AAA449E	Fqlngd	Lvhfqyxi	XX2784463429326809797294 26	Cras Vehicula Aliquet Industries

rmeTecnico 1 | Read Onl

Output

Action Output

#	Time	Action	Message	Duration / Fetch
2	21:01:55	CREATE VIEW InformeTecnico AS SELECT t.id AS id_transaction, u.name AS user_name, u...	0 row(s) affected	0.031 sec
3	21:01:55	SELECT * FROM InformeTecnico ORDER BY id_transaction DESC LIMIT 0, 2000	2000 row(s) returned	0.078 sec / 0.031 sec